

Quantum-circuit construction of the space-group symmetry-adaptation transform $Q : |k\rangle|I\rangle \rightarrow |i\rangle|a\rangle|c\rangle$

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What this is

A Qualtran resource-estimation bloq, `SymmetryAdaptationQROAM` (`src/integrations/qualtran/symmetry_adaptation_QROAM.py`), that realizes the whole-space-group Clebsch–Gordan / symmetry-adaptation transform Q of the ISDF grid: it maps the grid basis $|k\rangle|I\rangle$ (momentum k , interpolation point I) to the irrep-adapted basis $|i\rangle|a\rangle|c\rangle$ (irrep i , partner $a = (\kappa, \mu)$, copy c). It is a *resource model*: built from sizes only (data-free), it lays out the real sub-bloqs so Toffoli / Clifford / qubit counts follow from `QECGatesCost` / `QubitCount`. Diamond is the running example.

The structure it exploits

Q is block-diagonal in k ; the k -points fall into **families** (stars) under the point group (order $|P| = 48$); within a family the member blocks are *phased-permutation images* of the representative,

$$Q_{\text{perm}[g_j, I], c}^{(k_j)} = e^{-i k_j \cdot L_I(g_j)} Q_{I, c}^{(k_0)}, \quad k_j = g_j \cdot k_0,$$

and the representative $Q^{(k_0)}$ is itself block-diagonal over interpolation-point **orbits** under the little group, $Q^{(k_0)} = \bigoplus_{\text{orbits}} U_o$, each U_o a small $d \times d$ unitary, $d \leq |P|$. So the only independent unitary data is $\sum_{\text{fam}} \sum_{\text{orbit}} |o|^2$ (the “unique data”), which is essentially flat in n_k .

Precise mapping and register states

Q is the unitary on the grid space whose matrix elements are the space-group Clebsch–Gordan coefficients; on a computational grid basis state,

$$Q |k\rangle|I\rangle = \sum_{i, a, c} \langle i, a, c | Q |k, I\rangle |i\rangle|a\rangle|c\rangle.$$

Use the family factorization (block-diagonal in k)

$$Q^{(k)} = P(g_k) \Lambda(g_k, k) Q^{(f(k))},$$

with $f = f(k)$ the family, $g = g_k$ the coset op ($g \cdot k_0 = k$), $\kappa = \kappa(k)$ the star-member, $Q^{(f)}$ the family representative’s little-group reduction (block-diagonal over orbits), Λ the diagonal fiber phase $e^{-i k \cdot L_I(g)}$, and $P(g)$ the fiber permutation $I \mapsto \text{perm}[g, I]$. Writing the little-group-adapted fiber index as $I' = (\mu, \nu, c)$ (little-group irrep μ , partner $\nu = 1..d_\mu$, copy c), the map is

$$Q |k\rangle|I\rangle = \sum_{\mu, \nu, c} e^{-i k \cdot L_{\text{perm}[g, I']}(g)} [Q^{(f)}]_{(\mu \nu c), I} |i = (f, \mu)\rangle |a = (\kappa, \nu)\rangle |c\rangle$$

i.e. k 's information is re-encoded into i 's family label and a 's star-member κ , while I is reduced to (μ, ν, c) .

Registers. Input $|k\rangle$ ($\lceil \log_2 n_k \rceil$ qubits), $|I\rangle$ ($\lceil \log_2 n_{\text{IP}} \rceil$); output $|i\rangle, |a\rangle, |c\rangle$; work $|f\rangle$ (family), $|g\rangle$ (coset op, $\lceil \log_2 |P| \rceil$), and a b -bit phase-gradient $|\phi\rangle$ — all work returned to $|0\rangle$.

Register-state trace (the circuit applies $Q^{(k)} = P\Lambda Q^{(f)}$ to $|I\rangle$, so $Q^{(f)}$ acts first):

$$\begin{aligned}
& |k\rangle |I\rangle |0\rangle_f |0\rangle_g \xrightarrow{\text{(C) } k\text{-analysis}} |k\rangle |I\rangle |f(k)\rangle |g(k)\rangle \\
& \xrightarrow{\text{(A) apply } Q^{(f)} \text{ to } I, \text{ ctrl } f} |k\rangle \left(\sum_{I'} [Q^{(f)}]_{I', I} |I'\rangle \right) |f\rangle |g\rangle \\
& \xrightarrow{\text{(B) } \Lambda: \text{ load } L_{I'}(g), +k \cdot L \text{ into } \phi, \text{ erase } L} |k\rangle \left(\sum_{I'} e^{-ik \cdot L_{I'}(g)} [Q^{(f)}]_{I', I} |I'\rangle \right) |f\rangle |g\rangle \\
& \xrightarrow{\text{(B) } P(g): I' \mapsto \text{perm}[g, I']} |k\rangle \left(\sum_{I'} e^{-ik \cdot L_{I'}(g)} [Q^{(f)}]_{I', I} |\text{perm}[g, I']\rangle \right) |f\rangle |g\rangle = |k\rangle (Q^{(k)} |I\rangle) |f\rangle |g\rangle \\
& \xrightarrow{\text{(C) reindex } (k, I') \rightarrow (i, a, c), \text{ uncompute } f, g} \sum_{\mu, \nu, c} (\dots) |i = (f, \mu)\rangle |a = (\kappa, \nu)\rangle |c\rangle.
\end{aligned}$$

The phase-gradient $|\phi\rangle$ carries the accumulated $e^{-ik \cdot L}$ and returns to $|0\rangle$; f, g are functions of k and are uncomputed. The net effect is the boxed map above.

The three-stage circuit

- (A) **Little-group orbit-unitary synthesis (dominant).** Load only the per-family-representative orbit unitaries. Orbits are grouped by size; each distinct rounded size $N = 2^{\lceil \log_2 d \rceil}$ is one block-diagonal `BlockUnitaryInterferometerSynthesisQROAM` with $n_{\text{blocks}} = \# \text{orbits}$ of that size, addressed by (family, orbit) — **never by k** . Multiplexed nearest-neighbour beam-splitter layers + phase-gradient QROAM.
- (B) **Member reconstruction.** A single *select over the $|P|=48$ point ops*: apply $\text{perm}[g, I]$ on the I register, load the fold-back lattice vector $L_I(g)$ from the $|P| \times n_{\text{IP}}$ geometry table, and add $k \cdot L$ into a phase-gradient register (`AddIntoPhaseGrad`) to apply $e^{-ik \cdot L}$. All tables are $|P| \times n_{\text{IP}}$, independent of n_k .
- (C) **Reindex** $(k, I') \rightarrow (i, a, c)$. The I' -part is a `QROAMClean` over (family, I') (n_{IP} -sized); the $k \rightarrow (\text{family}, \text{coset op } g)$ map is fixed-width decision arithmetic, so the only k -touching widths are $\sim \log_2 n_k$.

Why the cost does not scale with n_k

The expensive part (A) loads only $\sum |o|^2$ numbers, addressed by (family, orbit); members are rebuilt in (B) by a select over the 48 *point ops* (a crystal constant, not n_k) plus an arithmetic $e^{-ik \cdot L}$ phase from $|P| \times n_{\text{IP}}$ geometry tables. Nothing in the dominant path is addressed by k . Concretely, from $2 \times 2 \times 2$ to $6 \times 6 \times 6$ the number of momenta grows $27\times$ but the dominant Toffoli grows only $\approx 1.2\times$ (it tracks $\sum |o|^2$, set by the few large high-symmetry orbit blocks). A naive per- k dense model ($n_k n_{\text{IP}}^2$) would grow $27\times$.

Toffoli minimization

Two levers, both applied: (i) the interferometer backend replaces the earlier recursive-CSD (naive $\sim N^2$) route — a $\sim 17\times$ per-block reduction; (ii) for each size group the QROAM **select-swap** split λ (`log_block_sizes`) is chosen by an *exact argmin of the real `get_cost_value`* Toffoli (applied to the phase-layer, final, and final-adjoint lookups). E.g. the dominant $N = 4$ group (648 blocks at $6 \times 6 \times 6$) drops $10,733 \rightarrow 3,180$ Toffoli at $\lambda = 8$, and the argmin correctly reverts to $\lambda = 1$ for the small $N = 32, 64$ tables where select-swap does not help.

Verified resource counts (diamond)

mesh	n_k	$\sum o ^2$	Toffoli	Clifford	qubits (peak)
$2 \times 2 \times 2$	8	6,800	20,417	3.1×10^7	≈ 186
$6 \times 6 \times 6$	216	12,912	25,708	5.7×10^7	≈ 485

Total Toffoli grows $1.26\times$ for a $27\times$ increase in n_k (dominant stage (A) $1.23\times$, $\geq 82\%$ of the total). Relative to the recursive-CSD backend the interferometer + select-swap route is $23.9\times$ ($2\times 2\times 2$) / $19.2\times$ ($6\times 6\times 6$) fewer Toffolis. Counts come from `get_cost_value(bloq, QECGatesCost()) / QubitCount()`; a closed-form `estimate_symmetry_adaptation_resources` matches the built bloq. (The peak-qubit figure is register footprint + peak child ancilla; `QubitCount` on the call-graph-only bloq returns the smaller max-over-children value 133/427.)

Files

`src/integrations/qualtran/symmetry_adaptation_QROAM.py` (bloq + estimator + demo), `tests/test_symmetry_adaptation_qroam.py` (10 tests), material sizes in `src/integrations/qualtran/diamond_symm_orbit_data.json`.